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The Rhetorical Appeals in Interaction Design: Decolonizing Design for People of Collectivist Culture

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This paper aims to propose a conceptual framework to understand factors that restrain the engagement of users with collectivist culture background to computer technology. Drawing on the intersectionality of Hartson's Interaction Cycle and the classical rhetoric concept of logos, ethos, and pathos, this paper will discuss briefly the problem of HCI in the City of Surabaya, Indonesia as a case study. The application of rhetorical lenses in this research will aid the understanding Lloyd Bitzer's "rhetorical situation" that is active in the interaction of human and the machine. Rhetoric will also assist the actualization of designer's role from a merely aesthetic executor to a designer who equally engages in the conversation of HCI research and development along with HCI researchers and behavioural scientists.

Interaction design; rhetoric; design theory; HCI

1. Introduction

In 1947-1948, a mathematician Alan Turing introduced the idea that the computer could exhibit the intelligent behaviour as humans do, by demonstrating a chess game with the machine as third player (Shah, 2013, p. 2). During an interview with BBC in 1950, Turing predicted that in fifty years a machine would be able to play games better than humans do, and be better at answering viva voce questions (Shah, 2013, p. 2). Turing's predictions were realized when the IBM super computer "Deep Blue" beat Grandmaster Gary Kasparov in a chess game, the IBM Watson won a Final Jeopardy game in 2011, and IBM's collaboration with music producer Alex da Kid in 2016 to create a song titled "Not Easy". In an interview in the Elvis Duran and the *Morning Show*, Da Kid saw IBM Watson as not the way to replace humans, but rather as a different way to get information and to be inspired. This emerging phenomenon denotes a significant change on human's relation to the computer technology, as the artificial intelligence is excelling to a 'humane' quality.

Human civilization is inevitably evolving towards a technology singularity; the merger of technology and human intelligence, that transcends Humanism, a natural form of human being, to Transhumanism, an accelerated form of humans enhanced by the sophisticated technology

(Cordeiro, 2014, p. 237). Hayles (1994, p. 444) explains that there are three waves in the development of cybernetics; Homeostasis (1945-1960), Self-Organizations (1960), and Virtuality (1972-present). Hayles suggests that a conceptual shift is happening in matters of the material changes in artifacts on all three waves. The Homeostasis period marks the foundation of cybernetics during which, “human, animal, and machines to be constituted through the common denominators of feedback loops, signal transmission, and goal-seeking behavior” (1994, p. 441). Self-Organizations is a wave where “...organisms respond to their environment in ways determined by their internal self-organizations” (Hayles, 1994, p. 442). In the period of Virtuality, there is a detachment from a physical embodiment into informational patterns. Furthermore, within the Virtuality wave, human civilization is heading towards a Post-human wave, where humans and intelligent machines build a symbiotic relationship called Intelligent Augmentation. It is in the post-human wave, that Howard Rheingold asserts, “humans will contribute to the partnership pattern of recognition, language capability, and understanding ambiguities; machines will contribute rapid calculation, massive memory storage, and rapid data retrieval (Rheingold in Hayles, 1994, p. 467).

In many developing countries, such as Indonesia, three waves of cybernetics are not happening periodically as Hayes describes in his three waves. Instead, a cybernetics development phase is occurring simultaneously yet uneven across the archipelago, as a result of imbalance infrastructure development, computer illiteracy, poverty, culture and local influences. There are some regions and isolated tribes in Indonesia still in the Homeostasis phase because the local and spiritual leaders forbid their community to use modern technology. One such community is the Baduy tribes, who prohibit the use of electricity in their Kanekes Village, an isolated area in West Java (Ardan, 2008, p. 118). Still, in larger and highly populated capital cities in Indonesia, like Jakarta and Surabaya, there is a movement towards Smart Cities that are able to enhance access and availability to public services through the application of information and communication technology (ICT); this is seen largely in health and public administration services. The Surabaya City Government has strong commitment to promote the use of e-kiosk—touch-screen computers that are available in public offices, in that the citizens of Surabaya can independently register birth, death, or relocation of a family member through the e-Lampid website, or make an appointment at the nearby health center through the *e-Health* website.

However, the reality is, many people are not yet receiving any benefit from the device, as they are not using it with certain groups in the community that are untrained and unfamiliar with computer technology. A situation where over one hundred of people are waiting in a long line in a government civic records office to be served while there are couple of computer station unused becomes prevalent. According to The Indonesian Ministry of Communication and Informatics, on the computer ownerships statistic in 2014 shows that 72 percent people in Indonesia do not own a computer (The Indonesian Ministry of Communication and Informatics 2014). Thus, this larger group do not have the basic computer skill in operating computer-based government services, leaving the information technology facilities mostly left unused. Although they have the confidence to operate smartphones, they have a degree of anxiety to operate a touchscreen computer. Indeed, it is a contrast to the statistics of computer ownerships in Northern America, Europe, and some parts in Asia in which is exceed 50 percent (Global Computer Ownerships 2015). However, this number only represents one fifth of the world’s total population. The HCI research has been focusing on this niche population and left the majority unexplored and unserved.

Cultural influences and traditions make for difficult implementation and wide-spread use of technologies, like the e-kiosk system in Indonesia. This is because the modern computerized system – with its strong influence of Western ways of knowing, communicating, and culture – is designed to facilitate independent and individual problem solving. People in some developing countries like those I mentioned, have a strong character of in-group collectivist, which means they are work in groups and are less individualistic (Zhao Fang, Nin Sheng & Collier, 2014). Person to person engagement is essential for communities like these, so the idea of Post-human, in which the existence of *human* (in this context is a city government officer as operator/ assistant) is replaced by the machine (an e-kiosk), challenges the nature of Indonesians as a collectivist culture. To some users in a collectivist culture, the rhetorical situation has already been activated through the surrounded negative impressions and anecdotes before the user has made a contact with a device. David Blakesley describes anecdotes as, “a form of ideological maintenance. They express or reaffirm belief, acts that have a curative effect and share some functions with propaganda” (2002, p. 97). Some possible anecdotes that circulate among potential users in Indonesia towards the e-kiosk would be that they are “complex and difficult to operate”, “confusing”, that they break down easily” and the notion that there is “no need to worry about using the computer, because there will be an officer who is willing to assist you”. These anecdotes prevent potential users to use the interactive system as it failed to give them trust. Local citizens would prefer to travel far and wait in line for hours to be served in civic records offices instead of doing it by themselves from home.

The inequality of technological advancement in communities without access to technology leaves a major gap in Human-Computer Interaction scholarship, especially in user experience research represents people and cultures from developing countries. The current knowledge of User Interaction/User Experience (UI/UX) design, which motored by Western computer engineers and scientists has failed to overcome the technology anxiety in the third-world countries. The dominant design of human interaction with computer technologies and programs may be based in studies of people in developed countries, but are still constructed through the theoretical lens of the Western culture. Teena M. Carnegie, in her paper *Interface as Exordium: The Rhetoric of Interactivity,* claimed that the features of the current technology such as images, video and audio can rectify face-to-face non-verbal cues in person-oriented environments (2009, p. 169). She also stresses that a social presence would not be necessary to the task completion in Human-Computer Interaction as a user would not be interested in engaging with other individuals while they were focusing on completing the required tasks (Carnegie, 2009, p. 170). This notion of independence is a reflection of Western’s individualist culture; a culture focusing on self-reliance and solitude. It may not apply to users who came from a non-Western world as it challenges the nature of collectivist culture which values reciprocity, dependence, and harmony.

This paper aims to propose a new framework to support the engagement of users in developing world with a collectivist culture background drawing on the intersectionality of the rhetoric and Human-Computer Interaction. The application of rhetorical lenses in this research will aid the understanding on what Lloyd Bitzer calls as a rhetorical situation; “a complex of persons, events, objects, and relations presenting an actual or potential exigence which can be completely or partially removed if discoursed, introduced into the situation, can so constrain human decision or action as to bring about the significant modification of the exigence” (1992, p. 6), that is active in the interaction of human and the machine. Rhetoric will also assist the actualization of designer’s role from a merely aesthetic executor to a designer who equally engages in the conversation of HCI research and development along with Human-Computer Interaction (HCI) researchers and behavioural scientists. This paper will dismantle the potentiality of Machine Rhetoric of computers in order to explore the possible ways that design can bridge the listening game between computer (machine) as a Speaker and users as a Listener. The first part of this essay will discuss Barry Brummet’s rhetorical potential of the current technology, followed by the rhetoric discourse of design affordances in machines as Ethos, understanding the notion of users as listeners in rhetorical situation, and the actualization of designers to engage in the listening game.

2. Rhetorical Potential of the Machines

Rhetoric is an art of persuasion that is practiced widely in public speaking and writing; examples include arguing legal cases in a court, journalism and advertising. Classical rhetoric was predominantly applied in speech, and later in written texts, visual arguments, objects, and audio-visual rhetoric. There are three rhetorical appeals; Logo (rational arguments or the message) Pathos— (appeals to emotions of the audience) and Ethos— (character and credibility of a speaker).

George Kennedy states that rhetoric is working as the possessed energy that drives the speaker to speak, the energy that is embodied in the utterance, the underlying energy in messages, and the energy exposed to the recipient in interpreting the message (Kennedy, 1992, p. 2). Barry Brummet explores the rhetorical potential of machines. It is manifest in the enjoyment of sensory and aesthetic experience of people towards the machines which can constitute further 'attitudes, actions and commitments' (1999, p. 2). Brummet's definitions on rhetoric of the machines places emphasis on Pathos: the emotion, sensations, and enjoyment of the machine aesthetics. He explains, "The rhetorical effects of machine aesthetics are embodied in the meanings and significations facilitated by difference aesthetic experiences. In other words, an aesthetic experience is meaningful, and that meaning lends itself to rhetorical manipulation" (Brummet, 1999, p. 3). He highlights the understanding of rhetoric of machines theory as grounded from a possibility of "a system objects that generates meanings" (1999, p. 27).

Brummet defines machine aesthetics in three types of dimensionality; 1) Mechtech, classical machine technology and industrial machines 2) Electrotech, the aesthetics of high-technology machines of present times, and 3) Chaotech, the aesthetic of the decayed machines (1999). Brummet argues that the rhetorical struggle of the meanings of machines lies beneath people's social and psychological reactions to technology, and is shaped by their interaction experience with the given technology. Ewen and Ewen's statement in Brummet, that machine era transcends hope and horror, ambiguity and confusion, contradicts the computer anxiety phenomenon in developing countries. Kennedy contends that a mental effort and expenditure of energy are required in order to respond effectively to rhetorical assertions (1992). Brummet also suggests that design and engineering could offer strategies to construct an intentional rhetoric of machines, and specific premises for a social change; in other words, to prepare users psychologically to decode the message.

The rhetoric of computers engages in two ways; the first way is the external skin, the outer shape of the computer, and the second way is the software, a cyberworld; the utopia of fantasies within the machine once the user interacts with the computer. Classified as electrotech, the computer and high-technology mobile device acquire cognitive power, in which the processes and effectiveness are working down to the human scale (Brummet, 1999, p. 59). The present electrotech, ranging from video game consoles, computer artificial intelligence, home assistants, tablets, virtual reality gadgets, and hands-free speakers have dematerialized computers beyond its solid forms and becomes the extension of the human as it is controlled by the human gestures of touch and voice. Martins and Emanuel see the opportunity of interaction as rhetorical studies to be as ubiquitous to the digital world, as the interactivity of digital technology allows users to become participants, experiencing the sensations of being active in the process that is situated on the screen (Martins & Emanuel, 2014, p.1). The rhetoric of interactive websites aimed to persuade users to purchase a product or a service, include users in a brand activation, as participants in an on-going project. However, Brummet notifies that the surface form of present electrotech is an enigma to its beginner users, evoking both excitement and anxiety towards its appearance and logic (1999, p.65). This situation makes a device unable to yield its function and either able to clearly demonstrates its features. Thus, the knowledge of the depth of a computer intensifies solely in the continuous engagement to the computer software.

3. Affordances as Ethos

The principles of Aristotelian persuasion have been adopted by design theorists and explored by design practitioners as tacit values reflected in their works. A designer is no longer subjected to styling and aesthetic value of outer case of an object. In fact, in specific design cases, today's designers are pushed to gain a high understanding of subject matter such as the development smart cars and smart homes, that draws on natural science, technology, culture, process and making, social sciences and engineering. To explain the intersectionality of design and rhetoric, Buchanan (2001) calibrates logos, ethos, and pathos in design realms. For the designer, logos is technological reasoning or intelligent structure of subject matter; pathos is affordance: the way a product must be understood by targeted users in making it function, and Ethos is close to "brand", a gut feeling--a manifested character or identity of a designer or manufacturer in a product or product series that comes from aesthetics or appealing shape of a product.

Affordances are not solely translated as pathos in the current electrotech, instead, as ethos relates to character and identity of an object, therefore, the four affordances will simultaneously generate ethos of an electrotech. The cognitive and physical affordances embodied in the surface of electrotech, are both pathos and ethos. The physicality of an electrotech must be able to deliver hints and clues on how to use the machine (pathos), and the experience while operating the machines will leave an impression thus will become a character reference of the electrotech (ethos). The logos and ethos also happen on cognitive, sensory, and functional affordances in the depth of electrotech, in which the logos is subjected to software and applications, content, messages, and interface design of the object, and the ethos will be left as the experience using the content of electrotech. In example, the brand of Apple used to be identified with Steve Jobs in which the ethos of Jobs (visionary and sophisticated design) embodied in its products. As ethos yields a character of electrotech, it therefore transforms a machine from a Non-Being to a Being, an object that brings a meaning to human.

The depth (software and operation system) and the surface (outer physical appearance) of present computer technology are of a different quality with boundaries and dualisms, which is encapsulated in the Interaction Design canon through affordances appeal. Donald Norman defines the term affordance by referring to Gibson's definition: "an attribute of an interaction design feature is what that feature offers the user, what it provides or furnishes" (Hartson, 2003, p. 316). Norman goes on to explain that Discoverability and Understanding are substantial in creating good design. Discoverability refers to what kinds of actions are possible, and where and how to perform them, while Understanding relates to how the product could be understood effortlessly in regards to its use of panels, settings, knobs, buttons, etc. Discoverability draws in five fundamental psychology concepts, affordance, signifiers, constraints, mappings, and feedback. Norman notes that affordance is a physical relationship between an object/artefact and the user that determines how the object could possibly be operated. According to Norman, there are two types of affordances; 1) Real Affordances: a physical appearance of an object, and 2) Perceived Affordances: characteristic in the appearance of the object that gives clue how to make the object be functioned (Hartson, 2003, p. 316).

Adapting Norman's model, Hartson (2003) details affordances in the context of Human-Computer Interaction (HCI) and usability engineering canon into four types of Affordances: 1) Cognitive affordance: "a design feature that helps, aids, support, facilitate, or enables thinking and/ or knowing about something", 2) Physical affordance: "a design feature that helps, aids, supports, facilitates, or enables physically doing something", 3) Sensory affordances "a design feature that helps, aids, supports, facilitates, or enables the user in sensing (e.g. seeing, hearing, feeling) something", helping users with sensory affordances and 4) Functional affordances: "design feature that helps users accomplish work (i.e., the usefulness of a system function) (Hartson, 2003, p.316-323).

In his book *Design of Everyday Things,* Don Norman also contends that experience is critical to interaction design, therefore the designers' objective is to produce pleasurable experience (1999, p. 10). His statement legitimizes Brummet's rhetoric of machine as pathos, which underpins the user's sensation, enjoyment, and emotional experiences during interaction with the machine. These sensory experience which extend beyond words are described by Kaufer and Butler as 'rhetorical situations' (1996, p. 34). Kaufer and Butler argue that rhetorical situation also involves "ancillary events", a preliminary milieu that aim to grab users' attention into the primary and sustained object. For example, a book cover or a packaging design are ancillary events to build awareness of targeted audience to the substantive object: the book content or the actual product inside the packaging.

Table 1. Hartson's (2003) Affordance Types

Affordance type	Description	Example
Cognitive affordance	Design feature that helps users in knowing something	A button label that helps users know what will happen if they click on it
Physical affordance	Design feature that helps users in doing a physical action in the interface	A button that is large enough so that users can click on it accurately
Sensory affordance	Design feature that helps users sense something (especially cognitive affordances and physical affordances)	A label font size large enough to read easily
Functional affordance	Design feature that helps users accomplish work (i.e., the usefulness of a system function)	The internal system ability to sort a series of numbers (invoked by users clicking on the Sort button)

Still, it is the way in which the user responds to the rhetorical situation that determines the success of the interaction design. In interaction design, there is a goal and purpose, therefore, the rhetorical situation must be able to both build users' awareness and facilitate them to accomplish the requested task. For example, the rhetorical situation in government e-kiosk for civil records must be able to motivate people to use it, and to direct people to successfully report birth, death or address change. Moreover, the rhetorical situation will be considered as a failure if the users are not engaged to with the technology, and thus will not be able to complete the task or simply be confused by the commands and the menu. Therefore, the user is an important element to the effective result of interaction design.

Table 2. Rhetoric appeals in Electrotech

Electrotech	Surface (Physical and Cognitive Affordances)	Depth (Cognitive, Sensory and Functional Affordances)
TV	Shape and Form (L, P, E) Screen size and resolution (P, E) Sound quality (P, E) Button (P, E) Brand Logo (L, E) Remote (P, E)	Interface design (L, E) Compatibility with other devices (L, E) Gesture commands (L, E)
Touch-screen / Non-touch-screen computers/ Tablets	Screen responsiveness (P, E) Screen resolution (P, E) Size (L, P, E) Casing material (L, P, E) Colour (L, P, E) Sounds (P, E) Brand Logo (L, E) Buttons Accessories (P, E)	Operating system (L, E) Software (L, E) Applications (L, E) Games (L, E) Compatibility with other devices (L, E)
Home assistant (Amazon Echo/ Google Home)	Size (P, E) Brand Logo (L, E) Colour (P, E)	Skills (L, E) Voice responsiveness (L, E) Compatibility with other devices (L, E) Smart home partnerships (E)
Rhetoric appeals: (L): Logos (P): Pathos (E): Ethos		

4. Users: The Other and the Listeners

The first section discussed the notion that the electrotech is an actualization of Western domination (developed countries) imposed upon other cultures (developing countries). This section will discuss user profile in a developing country (Indonesia) to understand the Otherness and Non-being boundaries and possibilities in building an effective rhetoric strategy. The Western culture is predominantly rooted in the Ancient-Greek culture, which is well-known for undermining and dictating other cultures, foisting race superiority and male-domination. The result if the roots and manifestations of Western cultures in the developing countries that used to be under colonialism of the European for centuries, was that those countries found themselves as an indigenous people, situated in the colony as the Other and Non-Being.

The occurring phenomenon of the machine dominance as the speaker simulates Scott's western rhetoric concept that isolates Listeners (citizens/users) as a passive element. Scott (1975, p. 442) asserts that a listener-oriented rhetoric is foreign to Western culture. Triandis (cited in Novera, 2004, p. 477) explains that "In individualist Western culture, people think more in terms of 'I' than 'we' and focus on the interests of themselves and their immediate family. In an individualist culture the cardinal values are creativity, bravery, self-reliance, and solitude. In collectivist culture, the values are reciprocity, obligation, duty, security, tradition, dependence, harmony, obedience to authority, equilibrium, and proper action". The Virtuality that Hayles speaks of previously in the beginning of this paper, is a reflection of the Individualist Western culture: self-reliance and solitude, thus it eliminates the identity of collectivist culture: reciprocity, dependence, and harmony. The collectivist culture is a high context culture, where the meaning of communication also conveys through subtle hints such as body gesture, facial expression, a tone of voice, words choices, etc. People in high context culture tend not to care about the information in printed materials. For example, a person in Indonesia will automatically ask others how to operate a vending machine instead of carefully studying the step-by-step instructions that came with the machine. Even when the person is able to complete the task independently, they would still require a justification from a human being whether they did the required task correctly to get the peace of mind (Calhoun et al., 2002, p.296). In this notion, the existence of electrotech (which is intended to be the embodiment of the government) impedes the interactions of the citizens as listeners/users, the government as a speaker, and the message as service.

Poulakos' Sophistic Definition of Rhetorics implies the listening game. *Kairos* (the opportune moment), and *to prepon* (the appropriate) are two elements when the speaker (*Rhetor)* is acknowledging listener's existence. To Sophists- the teacher of rhetoric-, *logos is* their medium, and *Logos* is not to speak, but that which is spoken. Poulakos explains, "If what is spoken is the result of a misreading on the part of the rhetor, it subsequently becomes obvious to us, even to him, that 'this was not the right thing to say'. If silence is called for and the response is speech we have a rhetor misspeaking to an audience not ready to listen, or not ready to listen to what he has to say, or ready to listen but no to the things he is saying" (1983, p. 41). An audience needs to be in a state where they are ready to listen, to receive the information hence delivering a message requires an appropriate time and the right message content; "the right thing must be said at the right time, inversely the right time becomes apparent precisely because the right thing has been spoken" (Poulakos, 1983, p. 42). Furthermore, according to Poulakos' definition of rhetorics, it is possible to position electrotech as a medium or a *Logos*. An electrotech has been spoken as "inhuman" to users, creates the technological anxiety simply because it is a new idea of "human". As the users were [become] more familiar with human-to-human interaction, it conditioned the responses and restrict the actions (Poulakos, 1983, p. 44).

5. Designer as a Sophist

This paper opened with Turing's idea of machine having the minds of humans, or to be precise, "Is it possible to figure out digital computers which would be successful in the game of imitation?" (Turing in Gimenes, 2015, p. 427). This idea is challenged by Searle's Chinese Room Argument, that a computer program is not necessarily able to make a computer think, regardless the advancement of the technology is. Don Norman also claims, machines are 'conceived, designed and constructed by humans', while human minds are complex, imaginative and creative, fulfilled with common sense that built from years of experience (Norman, 2013, p. 5). Humans behave insufficiently towards a machine because it is not designed by the understanding of human behaviour. An interface designer's locus is to facilitate communication between humans and the depth of the electrotech in a natural, effortlessly, and less depressing, despite complexe computer transactions.

Cultural conventions have a significant influence, primarily in the effectiveness of cognitive affordances, as they depend on the attributes of both the user and the artefact. In the perspective of user-centered design paradigm, cultural conventions can be pulled out into how to understand the user in terms of 'cognitive, affective or behavioral point of view, as well as the social, organizational, and cultural contexts in which users function' (Bowler et al., 2011, p. 723) since every user are unique. Electrotech is a rhetor, and users are listeners. When users interact with electrotech, they will figure out the way it operates (the Gulf of Execution), and later on will try to figure out what happened (the Gulf of Evaluation). Designers as a teacher (sophist) are helping users to be 'active', and to cross those two gulfs by a specific action (Norman, 2013, p. 38).

The 'bridge' that is proposed by Norman is known as stages-of-action model, and is an iterative activity of seven stages; goal, plan, specify, perform, perceive, interpret, and compare. Hartson merges Norman's stages of action with affordances concepts in usability engineering which he created the Interaction Cycle for User Action Framework (UAF). The Interaction cycle starts from Planning, Translation, Physical Actions, Outcomes, and Actions. Planning is about how interaction design direct user to know *what to do* with the artefact; Translation is how interaction design helps users to know *how to do,* translating task plans into specific actions; Physical Actions is how interaction design supports user *doing* the action; and Assessment is how well feedback in interaction design helps user to *assessing outcomes* of actions (Hartson, 2003, p. 329). Thus, this model will assist people from collectivist cultures to discover a better user experience design that fits to their needs.

The rhetorical appeals also emerge in the Interaction Cycle as shown in Figure 4. On Planning, a user will have first encounter to a medium (logos), where they will experience the existence of physical presentations such as sizes, colours, position and size of a button (affordance/ pathos), and also the identity of an artefact such as brand, logo of an institution/ company (ethos) will help them to translate what actions that should be executed. In the Translation phase, the interface design/ menu/ button (pathos) and commands (logos) in the artefact will help them to discover how to execute the specific actions. Physical actions are those that occur in the stage when the users do the required action with the help of button/ menu/ interface design (pathos) and commands (logos). Ethos is working along with the outcome as the impressions left felt by users during physical actions, and also during Assessment when users evaluate the outcomes with the feedback shown on the screen (pathos and logos).

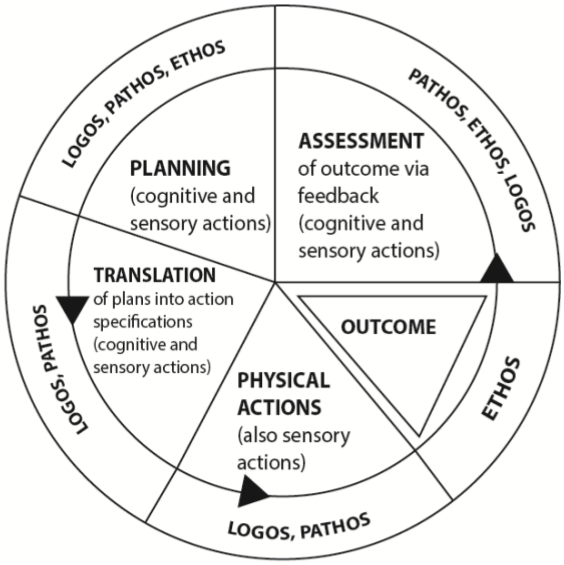

Figure 1. Rhetoric appeals in Interaction Cycle

There has been a growing interest in bringing the rhetorical canon to interaction design research, as the Field rhetoric plays a significant role in developing the discourse of experience design through application software analysis and development. Recent publications also suggest the implementation of a rhetorical lens in user experience will aid the discovery and the understanding of a design application software to a different level. There are three key concepts of experienced centered design that is adapted from a philosophical approach of John Dewey and Mikhail Bakhtin: 1) a holistic approach to experience, 2) continuous engagement and sense making, and 3) relational or dialogical approach (Salvo, 2001, 4). User/listener-oriented realm is also possible in the participatory design research with designers as the agent. In participatory design research, designers-researchers and the participants/ users are actively involved, iteratively formulating the problem solving. Participatory design research originated from Scandinavian countries in 1980s, when the computer technology was introduced to workplaces, and where the workers had limited knowledge towards computer technology (Spinuzzi, 2005, p. 164).

A rhetorician Kenneth Burke's work of identification can also be recognized as parallel to the user experience design canon. Different to Aristotelian rhetoric, in which rhetoric is defined as an art of speech and ways to gain advantage in an argument, to Burke, rhetoric is about identification. His method of Dramatism and traditional rhetoric both are frameworks that aim to understand the human relations that form through language (Burke, 1969). On David Blakesley's "the Element of Dramatism" (2002), identification is aligned with interests or motives, which allows to an unconscious part. Human's motives are always enigmatic; the sense of incomplete leads a self to seek for identification with others. Burke said that a self-identifies itself to an object (person or something), acquires a shared belief, desires to be a part of another, and galvanized by the commonalities. Individuals build their identities by relating themselves to other entities such as physical objects, values, idol, characters, community, groups, brands, and beliefs. The association formed between an individual with other object is what Burkes called by consubstantiality. Shared substance constitutes an identification between a person and some property of person: to identify A with B, is to make A consubstantial with B. However, in consubstantiality, a self still possesses their identity that differentiates them from an object they identified with. This divide brings value to each individual. Burke's theory of rhetoric has the potential to be a perpetuation of the third concept of experience design, because parties who are engaged in an experience gaining a shared value, which makes them consubstantial. The tools to reach this consubstantiality is by a common language.

Omar Sosa-Tzec claims that a rhetorical situation in a user interface is happening at the level of function, where the action is communicated by the visual artefact (2014). Sosa-Tzec argues that the visual artefact is a core element to such apps, the way its shaped and resonated to users controls the degree of effectiveness of an interface design, persuade people to choose them, to operate them hence accomplish the designated goals. He sees the relevance of information design in the persuasive interface design as it is constructed in information and interactions. He notes, “A persuasive interface partially depends on the way information relates to and is affected by the interactions with the system, and the way that both aid the user to make sense of the experience.” (2014, 1). Sosa-Tzec grounds his position by using Kenneth Burke’s rhetoric of identification and Aristotelian’s visual enthymemes. He understands HCI through Burkean dramatism rhetorics concept that is called as Pentad, as a justification for his argument that persuasion in interface design derives from identification. Burke’s Pentad uses five key elements; first is *Act* in which the action took place, second is *Scene*, the situation where the action takes place, third is *Agent*, the performer who enacted the act, forth is *Agency*, means and manner by which the agent performs the act, and the last one is *Purpose*, the motive of the performer’s act (Burke, 1969). Using the Pentad, Sosa-Tzec explains that the process of identification in interaction design is to establish paired relationships between designer, stakeholders, client and user. He also argues that a critical observation on Graphic User Interface (GUI) in a screen-based application is potential to formulate the possible motives to led the current design and then evaluate upon implications for the user experience. Sosa-Tzec adds that interfaces could also be regarded as interactive visual enthymemes produce a connection between interfaced and visual argumentation.

Zimmerman, Forlizzi, and Evenson of Carnegie Melon University claim that designers’ role in interaction research is merely that of a consultant that created pretty interface design, relegating them to the end of the interaction design research process (2007). This isolation creates a situation where the produced artefacts were not effective, since the designers came too late in the process. They further argue that design brings an added value to HCI research process. First, interaction designers provided a process to engage major under-constrained problems that were difficult to address by the traditional engineering approaches. Second, designers can integrate ideas from art, design, science and engineering in an attempt to create aesthetics yet functional interfaces. Lastly, designers engaged users in the process of developing the interface design by considering their needs and designers from the perspective of an external observer (Zimmerman, Forlizzi & Evenson, 2007, p. 5-6). Unlike engineers and business people, effective designers start their work by trying to understand the real issues, instead of trying to resolve a problem (Norman, 2013, p. 218). Designers are rhetorical thinkers, who focus on ‘conceiving, planning, and making products that serve human beings’ (Buchanan, 2001, p. 187). Buchanan claims designers are imperative in humanizing technology, bridging the interaction of human and computer, accommodating the diversity of human expectations underpinning in their backgrounds, culture, religion, origin, and living environment. Designers now concentrate on developing human systems which are found in their particular interactions and personal pathways.

6. CONCLUSIONS

The rhetoric of current high-technology machines or electrotech is defined by the enjoyment and excitement while having an interaction with it. This feeling will be the ethos of a machine, which works as image and identity similar to branding and people’s ‘gut feeling’ about an entity. The rhetorical appeals; logos, ethos and pathos work simultaneously within the interaction process between human and the machine, detailed in design affordances and interaction cycle. Computer anxiety happens because of an incorrect interpretation of ethos in the form of difficult, complex, easily broken machines, and this is something that is not trustworthy. The machine as a rhetor are the Other and the Non-Being to users; its language is difficult to understand and not easy to operate. The physicality of the machines does not provide adequate information to users, and it does not support to the users to know what to do, how to do, how to proceed the action and provide feedback to users. Further directions for this paper have been suggested concerning case studies towards users in developing countries or communities with limited access to technological advancements. The investigations on how these users perceive the Rhetorical appeals of electrotech will give significant contribution to interaction design and machine rhetoric scholarships.

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